

EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

Describe the fact-finding techniques, iteration plan, prioritisation and estimation methods used in the project.

1.1 Requirement acquisition technology(Fact-finding/elicitation)

The project needs are derived from a comparison of course assignment descriptions, QMPlus data, and stakeholder understanding.

First of all, we will make a structured reading of brief, clarify the mandatory functions and constraints (JSON persistence, no database), and build the role vocabulary (TA, Module Organiser, Admin).

Secondly, semi-structured interview and situational follow-up are adopted (see Section II Supporting material for the outline), and the scene and pain points are arranged and written into User Story (ProductBacklog_group013.xlsx). Rewrite the acceptance criteria in combination with prototype walk-through (Prototype_group013.pdf, UI design/) to cover the main path of registration-post-application-screening-employment.

Finally, the group workshop on fuzzy demand voting final draft, control scope spread.

1.2 Iterative planning (Iteration plan)

Iterations are consistent with "Iteration Sprint number" "Date started/finished" in Backlog: A total of four Sprint (approximately every two weeks) slices by dependency and demo increments.

Sprint 1: Deployable skeleton, home page offloading, registration and login, JSONs and deployment verification;

Sprint 2: File and skill labels, resume uploading, post

publishing and browsing; Sprint 3: Application and

status, screening by post, matching and short board,

matching algorithm;

Sprint 4: Employment/rejection, post closing, workload view, load balancing under equal matching.

Iteration	Planning cycle(Start - stop)	This iterative user story(Excerpt)
Sprint 1	2025-03-01 - 2025-03-14	US-001 Home page and role entry; US-TA-01 registration of candidates; US-TA-02 Login and logout of candidates; US-MO-01 Course organizer registration and login;
Sprint 2	2025-03-15 - 2025-03-28	US-TA-03 Maintain personal files and skill labels; US-TA-04 upload resume (plain text); US-TA-05 Browse to apply for position; US-MO-02 Publish Recruitment Position

Sprint 3	2025-03-29 - 2025-04-11	US-TA-06 Submit job application; US-TA-07 Query application status; US-MO-03 Screening candidates by position; US-MO-04 Check the skill matching degree and skill shortcut;
Sprint 4	2025-04-12 - 2025-04-25	US-MO-05 Employment or refusal of application; US-MO-06 Close the post; US-ADM-01 View the overall workload of the assistant; Load Equalization Recommendations for US-SYS-02 Equivalent Matching Degree

1.3 Prioritization(Prioritisation)

The Priority column of Backlog uses the MoSCoW variant

(Must/Should/Could). Current statistics: Must: 14items,

Should: 3items, Could: 1item.

Sorting basis: First deliver identity and persistence, and then open the end-to-end minimum path (position visible, application available, status verifiable). Management report and balance strategy are arranged in the Sprint after core closed-loop stability; Must is preferred when there is insufficient iterative content.

1.4 Estimation method(Estimation)

Estimation is listed as the story point (relative scale). The group adopts simplified Planning Poker (discrepancy value 3,5,8, etc.), and the

second round of convergence after the risk is explained by the high score when the divergence is large. Take US-TA-02 login, session and JSON read-write skeleton as reference (about 3–5 points); Items with complex verification or rules (such as publishing position and matching algorithm) across multiple pages tend to be 8 points. The plan will compare the story point totals with the available hours for the week of instruction, and move back Should/Could or split the acceptance criteria if exceeded; The review calibrates similar story points.

1.5 Technical description in requirements acquisition

Align with persistence mode(JSON, no database)

Brief requires that the JSON file be persisted and the database not be used. This will bring several hard constraints when obtaining requirements. It needs to be repeatedly confirmed in interviews and walks:

- Entity and relationship: The position, application, user/role, etc. can finally fall into the "object/array in one or several JSON files"; The interview asked "What job and TA information to associate with an application and how the status is transferred" is equivalent to clarifying the domain model and status machine.
- Concurrency and consistency (course-level simplification): Policies (e.g., single-process Tomcat, simple read and write) when multiple users write the same JSON at the same time should be explicitly assumed or downgraded at the requirement stage (e.g., Low Concurrency in Demo Environment) to avoid writing Bank-Level Transactions into the acceptance criteria.
- Traceability: In Backlog, the key Story can be marked with a persistent impact point (for example, "Field X Writing applications.json") to facilitate comparison with the score.

Align with Web Schema(Servlet/JSP. Sessions and Permissions)

During context tracing and prototype walk-through, the real path in the browser should be aligned with the server resources to reduce the gap of "There is no entry on the interface and back end":

- Authentication and Role: Whether TA/Module Organiser/Admin logs in by portal and how roles are distinguished in sessions will directly affect the permission acceptance criteria of Story such as "Who can perform post release/screening."
- Main path and URL resources: Registration-post-application-screening-employment response can be named example chain; During walk-through, you can write testable expectations into the acceptance criteria (for example, post-login jump, error prompt, 403/redirection behavior if required by the job) against "Each step is a form submission, redirection, or Ajax."

Lightweight Capture with Non-Functional Requirements (NFR)

When the brief is read in a structured manner, the technical NFR can be

recorded synchronously in addition to the functional list, for example:

- Deployment target (such as Tomcat + WAR), character encoding, basic security expectation (password display without clear text, session timeout, etc., subject to the scope of work);
- Maintainability: Data file location, readable JSON structure, so that the assistant can check whether the data is updated correctly.

These do not have to be written in lengthy specifications, but are listed in short tables in backlog or appendices and can be reviewed with Required Functions.

Demand for Technology Backfeeding: Prototype and Data Model Double Track

Prototype walk-through (Prototype_group013.pdf, UI design/) focuses on interaction and information architecture; The technical side can supplement the data perspective walk-through: For each key interface, ask "which record and which field will change in JSON after submission." If this is not correct, the requirement is still ambiguous on the field granularity or status enumeration and should be finalized back in the interview or workshop instead of leaving it to the code for guess.

2. Supporting material

Include supporting materials, such as interview questions, survey results, and customer feedback etc (if applicable).

2.1 Interview questions(Examples of Interview Questions)

- Please describe the actual steps and time consuming of your last application/assignment?
- What information (position, skill, status, contact) must be seen at a glance in the system?
- How would you like to roll back or correct a system error if you hired a candidate?
- What is the data backup and concurrency boundary you can accept when there is no database only JSON files?
- Administrator Perspective: How do I define "workload too high/too low"? What summary dimensions are required?

2.2 Survey results(Summary of survey results)

Lightweight Anonymous Questionnaire (5 points Likert plus open questions, small sample n=12 within the course).

2.3 Customer/user feedback(User feedback excerpt, alias)

MO-01: "Match score and short board are listed together, faster than just looking at the resume." TA-02: "Closed positions should not be able to apply."

ADM-01: "If you can see the list of jobs each person has hired, you will be more daring to adjust."
Mapped to corresponding User Story

Or Notes.

